X-Box , 2P, 4P & 6P
X-Box 2PS, 4PS & 6PS
X-Box 2PSN, 4PSN & 6PSN
X-Box 2PN, 4PN & 6PN

GENERAL INSTALLATION INSTRUCTIONS

CAUTION! Never Expose Eyes or Skin to UV-C – READ ALL MATERIALS BEFORE STARTING.



SAFETY CONSIDERATIONS

Improper installation, adjustment, alteration, service, maintenance, or use can cause fire, electrical shock, or other conditions which may cause personal injury or property damage. Consult a qualified installer, service agency, or your supplier for information or assistance. The qualified installer or agency must use factory kits or accessories when installing this product. Refer to the individual instructions packaged with kits or accessories when installing them. Follow all safety codes, wear safety glasses and work gloves. Read all instructions thoroughly and follow any warnings or cautions attached to any accessed area. Consult local building codes and the National Electrical Code (NEC) for all applicable requirements.

Understand the signal words **DANGER**, **WARNING** or **CAUTION**. These words are universally used for overall safety. **DANGER** identifies the most serious hazards, which will result in severe personal injury or death. **WARNING** signifies hazards, which could result in personal injury or death. **CAUTION** is used to identify unsafe practices, which would result in minor personal injury or product and property damage.



WARNING: Before installing or servicing this unit, turn off all power, there may be more than one (1) switch.

CAUTION-UV Exposure Precautions:



1. Doors and covers that give direct access to Ultraviolet (UV-C) lamp systems shall be equipped with an interlocking mechanism that removes power from the Ultraviolet (UV-C) lamp system.
2. It is recommended that access panels and components that are removable for cleaning and servicing and that provides access to other areas inside a unit be equipped with an interlock switch.
3. At some UV-C system installations locations, installers might require protective barriers (not supplied) and/ or PPE (Personal Protective Equipment) to protect installers & users (operational), service personnel and non-metallic parts from UV energy.

BARRIERS (if required, not supplied):

1. Barriers should be constructed of material that will not be affected by UV-C energy (e.g. metal).
2. Non-Metallic materials may be protected by directly attached barriers such as aluminum coated tape.
3. Both the unit inlet and outlet should be considered as possible Ultraviolet (UV-C) radiation paths when determining barrier locations.
4. The HAVC unit air filters are not considered Ultraviolet (UV-C) radiation barriers.
5. *Removable barriers shall be marked with the words "Caution" and the following or equivalent:* ***"UV LIGHT SOURCE - KEEP PROTECTIVE BARRIER IN PLACE"***

CAUTION-Material Exposure / degradation:

1. Materials subjected to irradiance from a UV-C lamp system shall be shielded from the UV-C light or be constructed of a material that is capable of withstanding expected UV-C exposure levels without degrading. Materials shall include, but not limited to, wiring, polymeric cabinets, structural components, gaskets, and filters.
2. Protective materials can include barriers (referenced above), conduit, aluminum tape, etc. Please reference ASHRAE® 2020-Systems and Equipment handbook for additional information[1].

[1] ASHRAE-2020. UV-C Photodegradation of Materials. Systems and Equipment Handbook. 17.6

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3. Inorganic materials such as glass, glass fibers, and metal are not affected by UV-C exposure[1][2][3].
4. Shield all organic material components within 5 ft **(**1.5 m) of the UV-C lamp[1]

WARNING: **Eye damage may** result from directly viewing the light produced by these lamps. To reduce the risk of exposure to UV-C radiation, take UV-C radiation protective measures for personnel during installation & servicing.

WARNING: This fixture is designed for use with germicidal UV-C lamps and must be installed in compliance with competent technical directions so that the user's eyes and bare skin will not be subjected to injurious rays.

WARNING: There shall not be any supply or return air openings or any other openings that are in direct line of sight of the UV lamp.

WARNING: There shall not be any openings in the duct that would emit light.

WARNING: UV Light Hazard. To Prevent exposure to ultraviolet light, be sure the ultraviolet air treatment system is disconnected before servicing any part of the HVAC system or removing any access panel or the equivalent.

CAUTION: Personal Injury Hazard. Power supply can cause electrical shock. Disconnect power supply before servicing or beginning installation.

CAUTION: Risk of exposure to excessive ultraviolet (UV-C) radiation - Do not operate without complete lamp enclosure in place or if sight lens is damaged

GENERAL:

1. RLM Xtreme™ lamps can be used to bathe surfaces, the air or both with UV-C energy. For surfaces locate the RLM Xtreme at known growth areas such as the coil and drain pan first, then other moist areas around humidifiers, etc. They can also be placed in return, supply and mixed air plenums or any combination of locations.
2. First check for power availability and probable service access.
3. Reflecting UV-C energy within a target cavity increases its fluence and overall effectiveness. Therefore, where possible, cover all likely reflection sites with UV-C reflective materials (e.g. aluminum tape) to reduce any shadowing and to increase the flux density of UV-C energy.

INCLUDED:

- RLM Xtreme X-Box "P" / "PS" / "PSN" / "PN" with
  - 120-277V power supplies (Qty. depends on X-Box size, e.g. 2, 4 or 6)
  - CU2™ Lamp/Ballast Sensor (Qty. depends on X-Box size, e.g. 2, 4 or 6)
- 1 ea RLM Xtreme IOM
- 1 ea Door Warning Label
- RLM Xtreme Installation Kit (Qty depends on X-Box size, e.g. 2, 4 or 6):
  - 2 ea Metal LampClamps
  - 8 ea Self-Drilling Mounting Screws for X-Box & LampClamps
- UV-C Lamp(s) sold separately
- **"PS" & "PSN" include 1 each 12V Magnetic Relay Switch with 20 ft Low Voltage Wire**

ELECTRICAL: LAMP WATTAGE/ POWER CONSUMPTION

Part Number	Description	Amps at 120V	Max °F	Electrical Watts*
55066559	SEL 33" Single Ended- High Output EncapsuLamp™	0.63	150	80
55064559	SEL 33" Single Ended High Output Lamp	0.63	150	80
55086559	SEL 61" Single Ended- High Output EncapsuLamp™	1.21	150	145
55084559	SEL 61" Single Ended High Output Lamp	1.21	150	145

*To calculate electrical amps, divide Electrical Watts by supplied voltage
X-Boxes include multiple variable input power supplies capable of 120-277Vac 50-60Hz, 1 PH, 1.25 -.54A. Calculate total amps necessary by multiplying each ballast's amp draw (depending on voltage).

[2] Kaufman, R.E. 2017. Study the HVAC System Photodegradation Caused by the Low Level UVC Light Irradiance Used for Coil Maintenance and Air Stream Disinfection. ASHRAE Research Project 1724, Final Report
[3] Kauffman, R. 2011. Study the degradation of typical HVAC materials, filters and components irradiated by UVC energy. ASHRAE Research Project RP-1509, Final Report.

INSTALLATION:

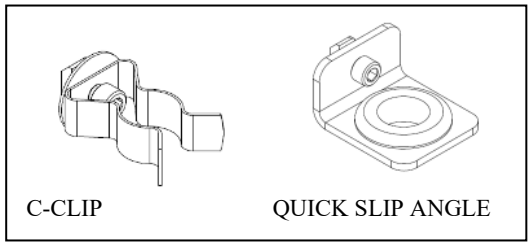
1. Consult all applicable codes before installing. Check fixture label(s) for the correct power requirements and supply the correct voltage from a suitable, protected (fused or breaker), and grounded power source.

CAUTION: **Using other than labeled voltages will void product warranty and may do additional damage to the entire system. Fixtures should be operated continuously to avoid growth of mold and bacteria under optimum conditions (AHU off).**

2. Power should be switched at all installation access points with a properly rated, SPST interlock switch (to completely de-energize the assembly when its install location is accessed. Accesses are to have signs in appropriate languages alerting maintenance personnel to the possible hazard of looking at or exposing skin to UVC energy (English label(s) provided, English/Spanish Plastic sign available separately). Signs must read "WARNING UV LIGHT SOURCE. DISCONECT POWER BEFORE SERVICING" any access may be equipped with a glass or Lexan® view port to view the lamps. Common glass or Lexan® will block (absorb) UV-C energy.
3. UV-C energy may damage other plastics, wires and other components. Protect suspect items with known UV-C resistant materials (e.g. aluminum tape) when they are in "line of sight" of UV-C energy.
4. Determine the most desirable UV-C lamp location for the intended use and then its power supply location. Note: The X-Box "P" and "PS" are NEMA 2/IP11 boxes recommended for indoor use – it may be mounted inside or outside of the target plenum (outside is preferred). If outdoors, the X-Box "PSN" is a NEMA 3R box that can be installed outdoors.

MOUNTING:

1. Use the spacing table below to determine the ideal distance between QuickSlip Mounting Kit (supplied). Mount using the provided T-nut & bolt. Lamp clamps may be mounted to any surface that addresses the intended site.

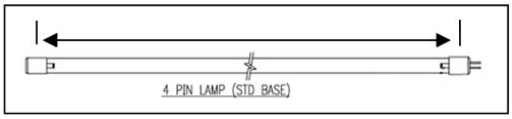


SAFETY NOTE: METAL C-CLIP SHOULD NOT BE ATTACHED TO THE "GLASS" PORTION OF THE LAMP.

LAMP SPACING TABLE

Nominal Lamp Length	Distance between Lamp clamps
33" (long) RLMX UVC Lamp	Approximately 31.5"
61" (long) RLMX UVC Lamp	Approximately 59.5"

Distance between LampClamps



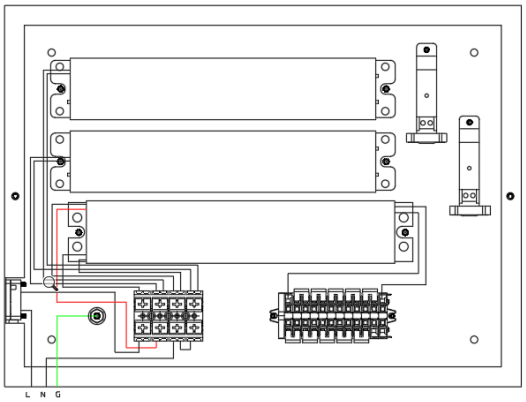
UV Resources X-Box™

Note: When choosing a mounting site, make sure that it can be supplied with the necessary power requirements, ballast hookup, and subsequent physical access for any future service.

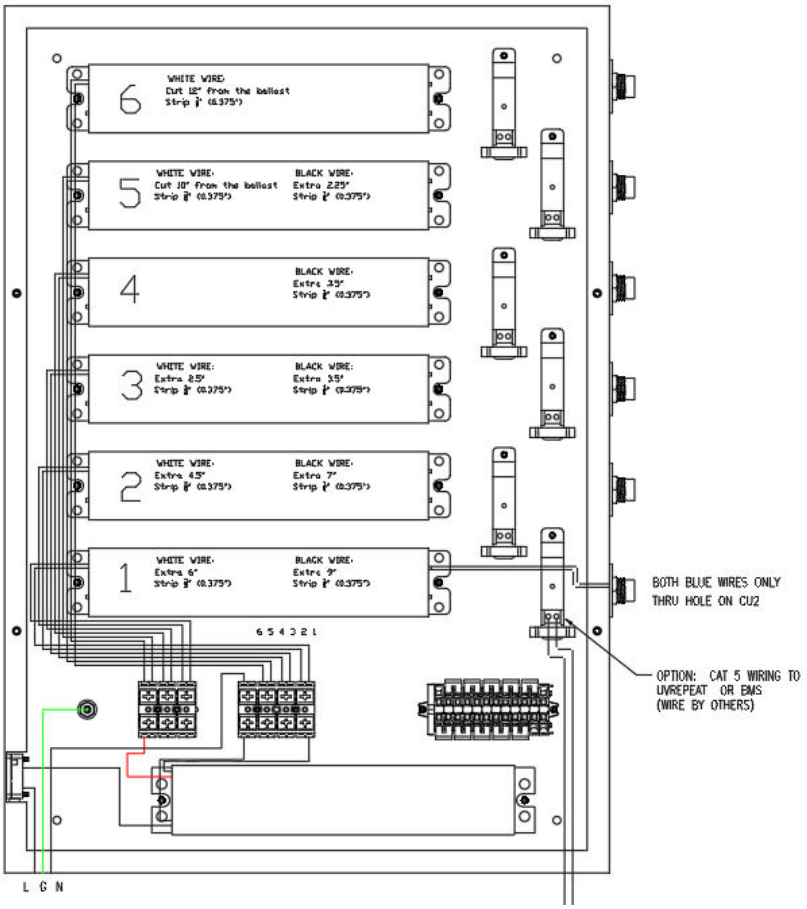
NOTE: X-Box 2PS, 4PS, 6PS & 4PSN Shown (X-Box "P" does not include 12V Contact Controller and Accessories).

VERSION VARIATION

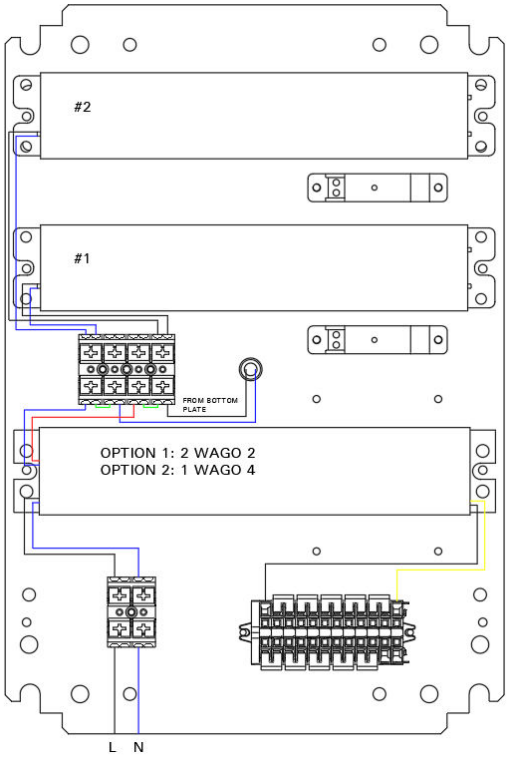
P POWER SUPPLY ONLY
PS POWER SUPPLY WITH INTERLOCK SWITCH
PSN POWER SUPPY WITH INTERLOCK SWITCH NEMA ENCLOSURE



2PS & WIRING



4PS & 6 PS WIRING



NEMA WIRING

Terminal Block Specifications

AWG	20-12 AVG
CONNECTION	CONTACT CAGE CLAMP CONNECTION
VOLTAGE EN	500V
SHOCK VOLTAGE	6 Kv
CURRENTY EN	32A
STRIP LENGTH	0.41 INCH (11 mm)

Dimensional Data & Capacities

X-Box "P" & "PS"

Size	W x H x D	Ballast Capacity	Terminal Block
1	14.75" x 5.75" x 1.75"	1	1
2	14.25" X 10.50" X 1.75"	2	1
4	15.00" X 21.50" X 1.75"	4	2
6	15.00" X 21.50" X 1.75"	6	2

X-Box "PSN"

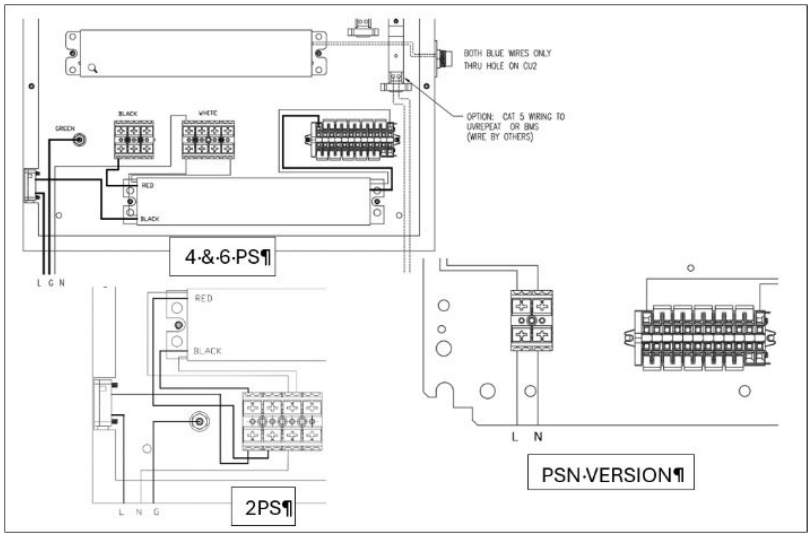
Size	W x H x D	Ballast Capacity	Terminal Block
2	15.75" 10.35" X 8"	2	1
4	15.75" 10.35" X 8"	4	4
6	15.75" 10.35" X 8"	6	4

POWER RATINGS:

THE RLM XTREME PRODUCTS COMES WITH 120-277 AC "UNIVERSAL" INPUT AT 50/60Hz POWER SUPPLIES.

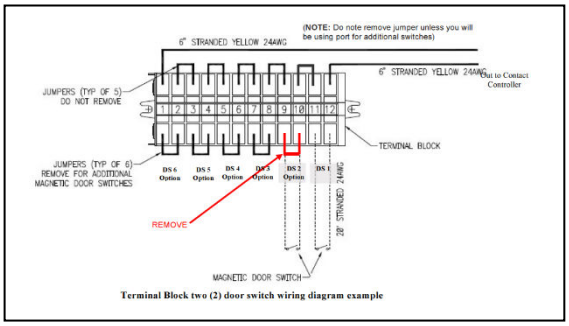
NOTE: Grounding: line-to-neutral. Consult local building codes and the National Electrical Code (NEC) for special requirements.

CAUTION: Always turn the system off before changing lamps. Failure to do so may produce a protection fault within the power supply. Before re-powering a fixture, follow manufactures instructions and make sure "all" connections (including lamp pins) are tight and complete to avoid sparks, shorts or overheating. A protection fault requires a reset procedure. To reset the "programmed start" function of a power supply, completely remove "all" power and power wires from the power supply and wait five minutes. Reinsert the power wires (L, N, G) into the power supply and reconnect A/C power. Then turn on all the switches. All lamps should be lit.



CONNECTING POWER:

1. Read fixture label power requirements – sum the power requirements for all assemblies associated with the install and install in accordance with National and local codes. Power should be connected to the X-Box via an appropriate conduit.
2. All power-in wire gauge and switch ratings must be in accordance with applicable codes. Power must also be switched at all installation access points with a properly rated, SPST interlock switch to completely de-energize the assembly when its install location is accessed (supplied with "X-Box PS"). Refer to 12V Contract Controller Diagram for PS models.
3. Refer to wiring diagram for wiring of ballast power.
4. Ground wire should be wired to the provided grounding lug.
5. Apply Door Warning Labels on all access doors/ports/panels as indicated above.



Terminal Block two (2) door switch wiring diagram example

RLM Xtreme IP67 Lamp Connector

RLM Xtreme Lamp Connector allows for the remote location of power supplies up to 24 feet (optional) from the UV-C lamp area. Connector should NOT be extended beyond the supplied 24 feet.

1. After the lamp and the X-Box location has been determined, measure the distance the LampLoom must span from the lamp to the X-Box. Cut the loom to proper length (Note: measure twice, cut once).
   Reminder: DO NOT extend the loom longer than 24 feet.
2. Remove the excess flex conduit (if supplied) and strip outer protective black jacket 3 ½".
3. Strip each wire (red and blue) ⅜" and twist stranded wires tightly.
4. Wires are now prepared to be installed into the power supply (see below).

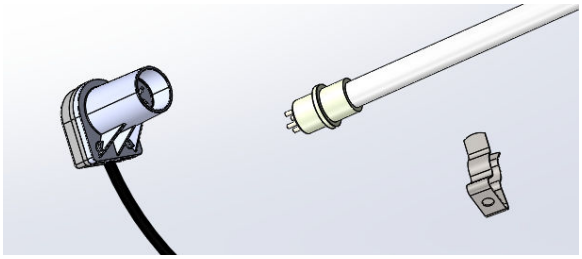
LAMPCONNECTOR CABLE COMES IN THE FOLLOWING OPTIONS & SIZE:

50000135-XX: M12 Extension Cable – 10' or 20' extension cable (optional) with Lamp Connector, cable strap kit, and lamp Quick Slip Kit.

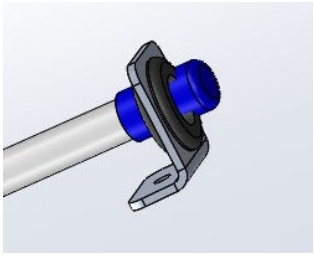
NOTE: RLM Lamp Connector from Lamp/Emitter to power supply may not exceed 24 feet. Extending the cable length beyond 24 feet will void warranty of system and may cause the system not to work properly.

INSTALLATION:

1. The Lamp Connector with Quick Slip Kit is designed for use with a 4-pin single ended T5 diameter lamp.
2. Once the proper location for mounting the lamp is determined and the mounting brackets are secured, attach the LampHolder and route loom to the power supply (see wiring diagram for proper wiring of loom to power supply).

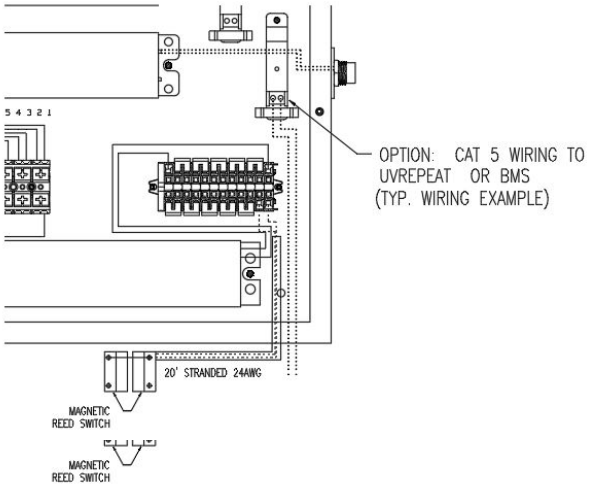


LAMP CONNECTOR LAMP



LAMPCLAMP QUICK SLIP

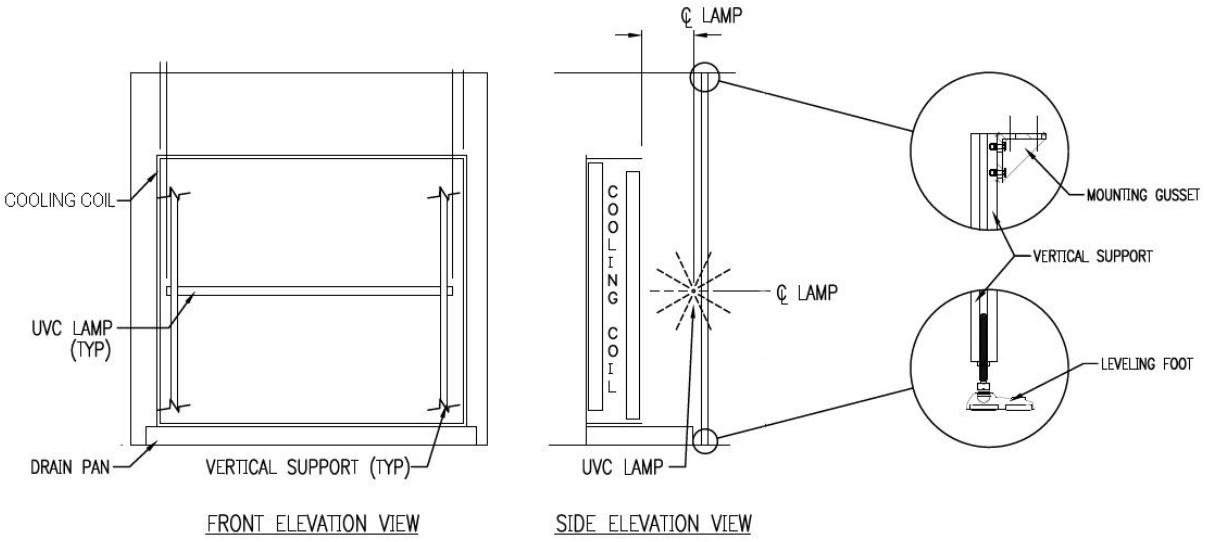
3. If BMS or UV COM (UV Repeat) connection is desired, connect wires as desired by the UV COM IOM.



LAMPS:

1. The RLM Xtreme comes with EncapsuLamp™, an FEP coating for lamp protection and for encapsulation should the lamp break. **DO NOT REMOVED THE FEP COATING FROM THE LAMP**.
2. Install the lamp into the LampHolder by:
   a. Loosening the compression nut.
   b. Slide lamp into holder, twisting it gently and slowly until the lamp pins align with the female plugs in the holder (Note, pins and plugs must align).
   c. Gently, yet firmly push the lamp into the socket by grasping the end of the lamp (not the glass) with the thumb and index finger. You should feel a slight "click" when the lamp is inserted correctly (**NOTE: IF LAMP IS NOT FULLY INSERTED INTO SOCKECT ARCING CAN OCCUR CAUSING FAILURE**).
   d. Slide Compression Nut over lamp and FINGER tighten to the LampHolder (do not over tighten).
3. Lamps should be replaced annually (8760 hrs.). Consider using UV-C Lamp Hour Meter (P/N: 90001203) to monitor lamp hours.

RLM XTREME AHU INSTALLATION DRAWING EXAMPLE



61” Vertical Support Centerlines Not to Exceed 59.5”
33” Vertical Support Centerlines Not to Exceed 31.5”

REPLACEMENT LAMPS	
Part Number	**Description**
55066554	SEL 33" Single Ended- High Output EncapsuLamp™ - 4 pack Includes FOUR each
55066555	SEL 33" Single Ended- High Output EncapsuLamp™ - 30 pack Includes THIRTY each
55086554	SEL 61" Single Ended- High Output EncapsuLamp™ - 4 pack Includes FOUR each
55086555	SEL 61" Single Ended- High Output EncapsuLamp™ - 30 pack Includes THIRTY each

RLMX Xtreme Accessories/Replacement Parts	
Part Number	**Description**
55002005	UV-C Fixture Framing Installation Kit (includes 2ea ~5' Struts, 1ea Gusset Kit w/allen wrenches, 1ea Leveling Foot Kit, 2ea splice)
90003000	Gusset Kit (used with UV-C Fixture Installation Kit).
90003010	Leveling Foot Kit (used with UV-C Fixture Framing Installation Kit)
90004044	CU2TM Ballast Sensing Monitor - 4 Pack
90004052	UVRepeatTM Kit 2 Board (Includes NEMA 4 Box & UV Repeat)- up to 15 CU2s
90004054	UVRepeatTM Kit 4 Board (Includes NEMA 4 Box & UV Repeat)- up to 28 CU2s
90001203	UV-C Hour Meter
90004315	12V Magnetic Relay Switch with 20ft Low Voltage Wire
90001865	LampLube™ - 5 Pack (Dielectric Grease- 5 gram packs)
50000137-04	IP67 LampConnector Cable Kit - Standard 4 Pin Lamp "UV Resources"
15000165	Quick Slip/Spring Clip Mounting Kit
55000054	Cable Strap Kit
50000135-XX	M12 Extension Cable - 10 Ft / 20 Ft

MAINTENANCE:

The intensity of the ultraviolet energy emitted from the UV lamps is dependent on the cleanliness and lamp age. The surface of the lamp should be kept as clean as possible for optimum intensity. Depending on the filtration level of the HVAC system and the general hygiene of the building, periodic cleaning may be necessary. Before attempting any maintenance procedures, always follow all warnings and cautions as detailed in this maintenance section.

CLEANING LAMPS

Note: If lamps are found to be broken, see the proper warning and cautions below regarding broken lamps and hazardous vapors.

1. Disconnect all electrical power to the unit and the UV lamps.
2. Wearing soft cloth gloves and safety glasses grasp lamp with fingers and twist lamp until it can be removed from tombstone.
3. Wipe down each lamp with a clean cloth and isopropyl alcohol. Avoid touching lamp glass with hands as skin oils can accelerate lamp degradation. (If lamps are coated with Teflon they can be touched with bare hands).
4. Apply dielectric grease to 4 pins. Place the lamp back in the tombstone and twist the lamp until it clicks into place.

REPLACING THE LAMPS ANNUALLY

Ultraviolet lamps should be replaced annually if operated continuously or after 9,000 hours of use if operated intermittently. Replacement lamps must be the specific size and wattage as originally supplied by the factory.

Note: Although the lamps may continue to generate a characteristic blue glow beyond 9,000 operating hours, the ultraviolet energy emitted by the bulbs degrades over time and will no longer provide the intended benefit.

LAMP DISPOSAL:

UV lamps should be treated the same as other mercury-containing devices, such as fluorescent bulbs, according to local regulations. Most lamps must be treated as hazardous waste and cannot be discarded with regular waste. Low-mercury bulbs often can be discarded as regular waste; however, some states and local jurisdictions classify these lamps as hazardous waste. The U.S. EPA's universal waste regulations allow users to treat mercury lamps as regular waste for the purpose of transporting to a recycling facility. The National Electrical Manufacturers Association (NEMA) maintains a list of companies claiming to recycle or handle used mercury lamps at www.lamprecycle.org.

LIMITED EQUIPMENT WARRANTY

FOR DETAILED INFORMATION, VISIT https://uvresources/warranty

The warranty (https://uvresources/warranty) supersedes and replaces any warranty statements orally made by a Salesperson, Distributor or Dealer or contained in the written instructions or other Brochures or informational documents in relation to this product. This warranty gives you specific legal rights, and you may also have other rights which vary from State to State.

UV Resources (UVR) warrants to the original buyer that its Products shall be free from defects in material or workmanship under normal use for period of one year. This warranty is contingent upon proper use of Products and will not apply if adjustment, repair or parts replacement is required because of an accident, unusual physical, electrical or electro-mechanical stress, neglect, misuse, failure of electric power, humidity control, transportation, unauthorized repair actions, or not installed or maintained in accordance with UVRs' specifications. This warranty is limited to the repair and/or replacement of parts. This warranty does not cover any labor or subsequent damage incurred as the result of Product failure or indirectly arising from the design, construction, installation, servicing, or operation of Products. UVR neither assumes, nor authorizes any person to assume for it, any obligation in connection with the Products. Buyers shall not return to UVR any allegedly defective goods without UVRs' prior written authorization. This warranty may not be assigned or transferred.

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